Research Paper

Customer Retention Strategy Using Engagement and Product Utilization Analytics

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Course: Data Analytics

Abstract—

Customer retention is a critical determinant of long-term business success and profitability. Organizations increasingly rely on data-driven strategies to understand customer behavior and reduce churn. This study presents a behavioral analytics approach to customer retention using engagement and product utilization metrics. An interactive dashboard was developed using Streamlit to analyze customer data, compute key performance indicators (KPIs), and identify high-risk customers.
The analysis reveals that customer engagement and multi-product usage significantly reduce churn probability, whereas high-balance inactive customers represent the highest financial risk. The study demonstrates how combining analytics and visualization can support proactive decision-making and improve customer retention strategies.

I. INTRODUCTION

Customer retention has emerged as a key strategic priority for businesses operating in highly competitive markets. Acquiring new customers is significantly more expensive than retaining existing ones, making churn reduction a critical focus area. High churn rates not only impact revenue but also reduce customer lifetime value and brand loyalty.

Traditional approaches to churn analysis often rely on demographic variables such as age, location, and income. However, these variables fail to capture dynamic customer behavior. This research emphasizes behavioral analytics, focusing on engagement levels, product usage patterns, and financial activity.

The objective of this study is to analyze customer churn using data-driven techniques and develop an interactive dashboard that enables organizations to identify at-risk customers and take proactive measures.

II. LITERATURE REVIEW

Customer retention has been extensively studied in marketing and data analytics literature. Kotler [1] emphasizes the importance of customer relationship management in improving retention. Similarly, research published in Harvard Business Review [2] suggests that a small increase in retention rates can lead to significant profit growth.

Recent studies highlight that behavioral data is more effective than demographic data in predicting churn. Engagement metrics, such as frequency of interaction and product usage, provide deeper insights into customer loyalty. Additionally, the use of dashboards and visualization tools has improved the ability of organizations to monitor customer behavior in real time.

The integration of analytics and visualization is therefore essential for effective decision-making in customer retention strategies.

III. PROBLEM STATEMENT

Organizations face several challenges in managing customer retention:

- Difficulty in identifying customers who are likely to churn
- Lack of understanding of behavioral factors influencing churn
- Inability to prioritize high-value customers
- Limited access to real-time insights

Existing systems often fail to provide actionable insights, leading to reactive rather than proactive decision-making. Therefore, there is a need for an analytical framework that integrates behavioral data with visualization tools to support effective retention strategies.

IV. OBJECTIVES

The primary objectives of this research are:

1. To analyze customer churn using engagement and product utilization metrics
2. To develop an interactive dashboard for data visualization
3. To identify high-value customers at risk of churn
4. To generate actionable insights and business recommendations
5. To demonstrate the effectiveness of behavioral analytics in retention strategies

V. DATASET DESCRIPTION

The dataset used in this study consists of customer-level information, including:

- Age
- Geography
- Credit Score
- Account Balance
- Number of Products
- Active Membership Status
- Estimated Salary
- Churn Indicator (Exited)

Data Preprocessing:

- Removal of irrelevant attributes

- Handling missing values
- Encoding categorical variables
- Filtering based on user-defined parameters

The dataset enables both descriptive and behavioral analysis of customer churn.

VI. METHODOLOGY

The methodology adopted in this study involves multiple stages:

A. Data Filtering

Interactive filters were applied to analyze customer segments based on:

- Age
- Balance
- Number of products

B. KPI Calculation

Key performance indicators were computed to summarize customer behavior:

- Churn Rate
- Active Users Percentage
- Average Balance
- Average Products per Customer
- Retention Strength Index

C. Customer Segmentation

Customers were categorized into segments:

- High Value Active
- Moderate
- High Value At Risk
- Low Engagement

D. Risk Identification

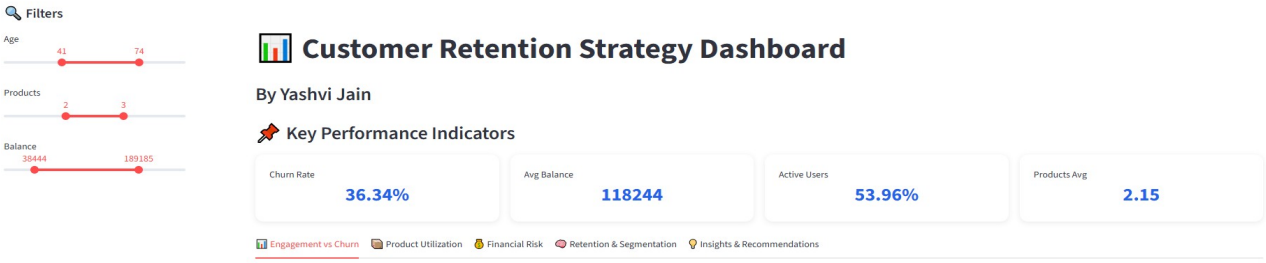
High-risk customers were identified based on:

- High account balance
- Inactive status

E. Visualization

Data was visualized using bar charts, histograms, and tables within the Streamlit dashboard.

VII. DASHBOARD DESIGN AND IMPLEMENTATION



(Fig.1. Customer Retention Strategy Dashboard Overview)

An interactive dashboard was developed using Streamlit to provide a comprehensive view of customer behavior.

Features:

- Multi-tab interface
- Interactive filters
- KPI indicators
- Visual analytics

Modules:

- Engagement vs Churn
- Product Utilization
- Financial Risk Analysis
- Customer Segmentation
- Insights and Recommendations

The dashboard enables users to explore data dynamically and derive insights in real time.

VIII. KEY PERFORMANCE INDICATORS

The following KPIs were used to evaluate customer retention:

1. Churn Rate
Percentage of customers who exited the service.

2. Active Users Percentage
Measures customer engagement.

3. Average Balance
Indicates financial value.

4. Products per Customer
Reflects product utilization.

5. Retention Strength Index
A composite metric combining engagement and product usage.

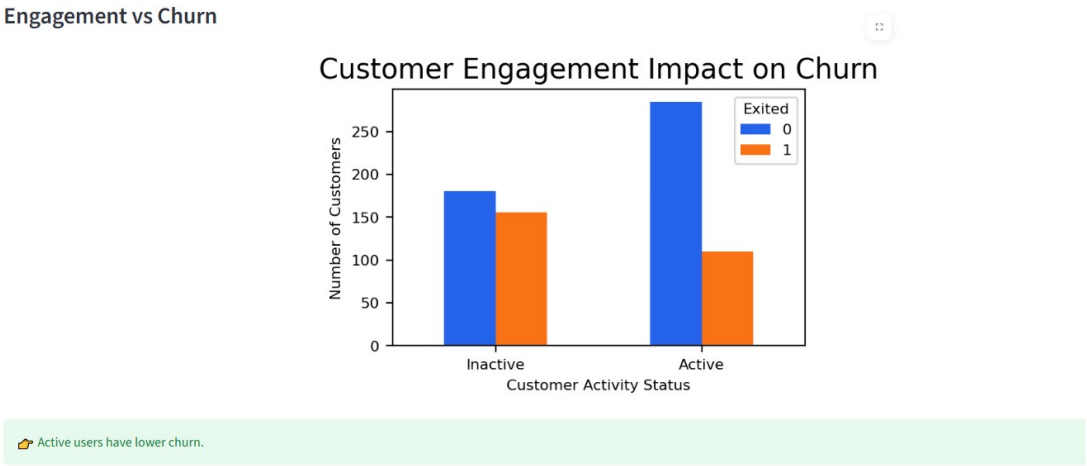
IX. RESULTS AND ANALYSIS

A. Engagement vs Churn

The analysis shows that inactive customers have a significantly higher churn rate compared to active customers.

Interpretation:
Customer engagement is a strong predictor of retention. Increasing engagement can significantly reduce churn.

Fig.2 illustrates the relationship between customer activity status and churn behavior.



(Fig.2. Customer Engagement Impact on Churn)

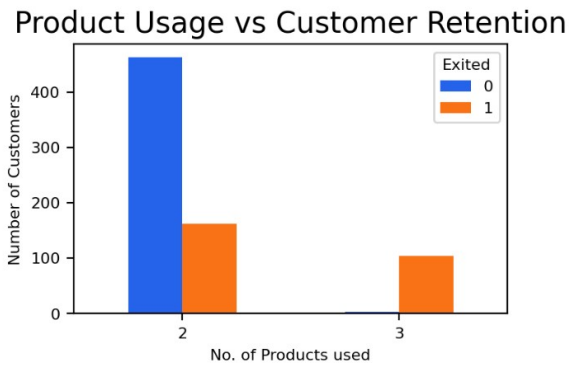
B. Product Utilization Analysis

Customers using multiple products exhibit lower churn rates compared to single-product users.

Interpretation:
Higher product adoption increases customer dependency and loyalty.

Fig.3 shows how the number of products used influences customer retention.

Product Utilization



More products = better retention.

(Fig.3. Product Usage vs Customer Retention)

C. Financial Risk Analysis

High-balance inactive customers are identified as the most critical risk segment.

Interpretation:
These customers have high financial value but low engagement, making them highly likely to churn.

Fig.4 highlights customers with high balances but low engagement , representing critical churn rate.

High-Value Disengaged Customer Analysis

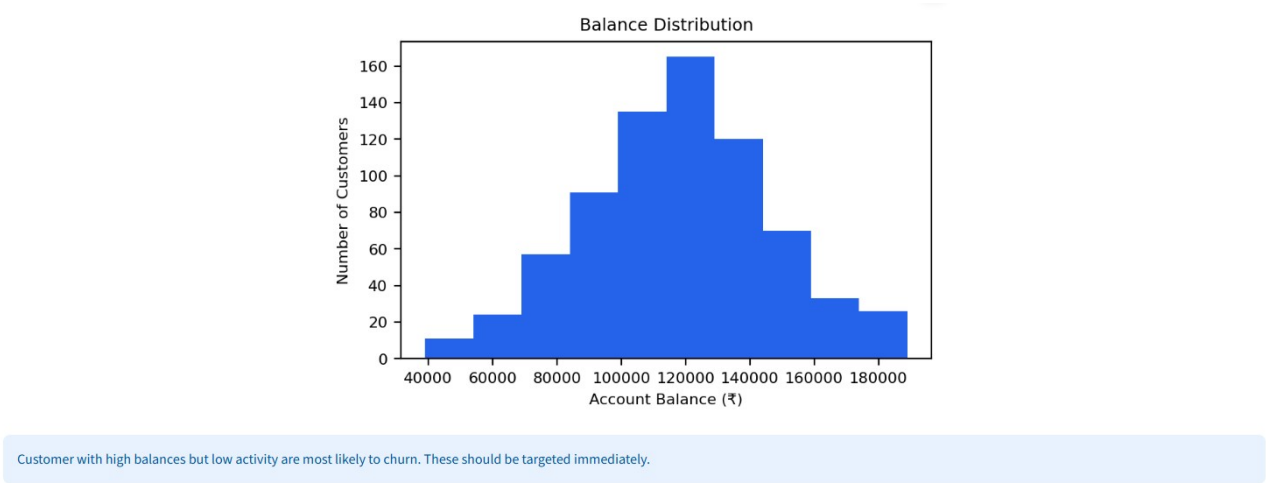
	Year	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
2	2025	15619304	Onio	502	France	Female	42	8	159660.8	3	1	0	113931.57	1
79	2025	15803136	Postle	416	Germany	Female	41	10	122189.66	2	1	0	98301.61	0
91	2025	15731511	Ritchie	808	France	Male	45	7	118626.55	2	1	0	147132.46	0
182	2025	15621075	Ogbonnaya	778	Germany	Female	45	1	162150.42	2	1	0	174531.27	0
392	2025	15573456	Cunningham	648	Spain	Male	46	9	127209	2	1	0	77405.95	1
449	2025	15658169	Cook	778	Spain	Female	47	6	127299.34	2	1	0	124694.99	0
584	2025	15803689	Begum	647	Germany	Female	51	1	119741.77	2	0	0	54954.51	1
589	2025	15591047	Ma	519	Spain	Female	47	6	157296.02	2	0	0	147278.43	1
670	2025	15576368	Bledsoe	624	Germany	Female	48	3	122388.38	2	0	0	30020.09	0
733	2025	15744607	Martin	738	Germany	Male	43	9	121152.05	2	1	0	64166.7	1

High Risk %

22.68%

(Fig.4. High-Value Disengaged Customer Analysis)

Fig.5 presents the distribution of customer balances used to identify high-value segments.



(Fig.5. Distribution of Customer Account Balances)

D. Customer Segmentation

Customers were segmented into distinct groups based on engagement and financial value.

Interpretation:
Segmentation enables targeted retention strategies and efficient resource allocation.

Fig.6 shows classification of customers into different segments based on value and engagement.

(Fig.6. Customer Segments Distribution)

X. DISCUSSION

The results demonstrate that behavioral metrics provide deeper insights into customer retention than traditional demographic variables. Engagement and product utilization play a critical role in determining customer loyalty.

Organizations can leverage these insights to design targeted interventions and improve customer experience. The use of dashboards enhances decision-making by providing real-time visibility into customer behavior.

XI. BUSINESS INSIGHTS

- Engagement is the strongest driver of retention
- Customers with fewer products are more likely to churn
- High-value inactive customers require immediate attention
- Product diversification improves customer loyalty

XII. RECOMMENDATIONS

Based on the analysis, the following recommendations are proposed:

1. Target high-value inactive customers with personalized offers
2. Encourage customers to use multiple products
3. Implement loyalty and engagement programs
4. Monitor customer behavior continuously
5. Use analytics dashboards for decision-making

XIII. FUTURE SCOPE

Future enhancements may include:

- Machine learning models for churn prediction
- Real-time data integration
- Personalized recommendation systems
- Sentiment analysis using customer feedback

XIV. CONCLUSION

This study highlights the importance of behavioral analytics in customer retention. By focusing on engagement and product utilization, businesses can proactively identify at-risk customers and implement effective retention strategies. The developed dashboard serves as a powerful tool for data-driven decision-making.

REFERENCES

[1] P. Kotler, Marketing Management, 15th ed. Pearson, 2016.
[2] Harvard Business Review, “The Value of Keeping the Right Customers,” 2014.
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